

CHM111 Practice Exam 2 on chapter(s): 3, 4

1. 25.47 mL
2. $2 \text{ Rb (s)} + 2 \text{ H}_2\text{O (l)} \rightarrow \text{H}_2 \text{ (g)} + 2 \text{ RbOH (aq)}$
3. 9
4. No reaction occurs.
5. 215 kg
6. 79%
7. $3.81 \times 10^1 \text{ g}$
8. 48.2%
9. $2 \text{ ZnS} + 2 \text{ C} + 3 \text{ O}_2 \rightarrow 2 \text{ CO} + 2 \text{ SO}_2 + 2 \text{ Zn}$
10. 2.501 g
11. $56.08 \frac{\text{g}}{\text{mol}}$
12. $\text{S}_8 + 12 \text{ O}_2 + 8 \text{ H}_2\text{O} \rightarrow 8 \text{ H}_2\text{SO}_4$
13. 0.208 M
14. 0.0817 M
15. 0.01659 g
16. $17 \text{ O}_2 + 2 \text{ C}_6\text{H}_{12}\text{O} \rightarrow 12 \text{ CO}_2 + 12 \text{ H}_2\text{O}$
17. Single-Displacement
18. $6.68 \times 10^{-33} \text{ moles}$
19. 5.120 g
20. 7.55 g
21. 0.06156 g
22. 8.455 g
23. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
24. 126 g/mol
25. 34.3 mL
26. neither oxidized nor reduced
27. $11 \frac{\text{mol}}{\text{L}}$
28. 0.6121 moles
29. $1.3 \times 10^{24} \text{ moles}$



31. molecular, acid

32. 265.3 mL

33. 2.682×10^{24} g

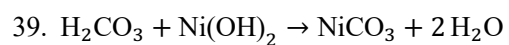
34. Oxidized: K Reduced: Na

35. Rb_2O

36. 1.86×10^1 mol

37. 4.020×10^{18} carbon atoms

38. 13%



40. Oxidizing Agent: H_2 Reducing Agent: K